NAME: ____

CENTRAL EAST EDUCATION DIVISION

2025 MSCE MOCK EXAMINATIONS

ADDITIONAL MATHEMATICS

Wednesday, 19 March **Subject Number: M132/I**

Time allowed: 2 hrs 30 min
01:00 – 03:30 pm

PAPER I

(100 marks)

Instructions

1. **This paper contains 12 printed pages. Please check.**
2. Before beginning to answer these questions write your **Full Name** in the spaces provided at the top of every page of the question paper.
3. Answer **all** the **seven** questions in **section A** and any **two** from **section B** the spaces provided.
4. **Section A** carries 60 marks and **Section B** carries 40 marks
5. Use of electronic calculators is allowed
6. All answers should be written in the space provided after every question.
7. **All necessary working should be shown** and any numerical expression being evaluated by calculators must be clearly stated; otherwise marks for the method may be lost.
8. The **final** answer to a question requiring the use of tables or calculators should normally be given to three significant figures.
9. In the table provided on this page, **tick** against the number of the question you have answered.
10. The marks allocated to question (or part of a question) are shown in brackets at the end of each question (or part of a question).
11. For **questions number 9(b) and 10(b)** graph papers will be provided. The spaces provided below the questions may be used for tables of values and answers.

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Section A (60marks)

Answer all the seven questions in this section

1. (a) Given that $f: x \to \frac{x+3}{x+2}, x \neq 0$, find $f^{-1}(-4)$ **(5marks)**

(b) Evaluate $\lim_{x \to 1} \frac{x^2-3x+2}{x-1}$ **(4marks)**

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2. (a) **Given that** $(\mathbf{1}+\mathbf{kx})^{\mathbf{n}}=\mathbf{1}+\frac{\mathbf{7}}{\mathbf{2}}\mathbf{x}+\mathbf{Bx^2}+\mathbf{Bx^3}+\cdots$ **where B and k are nonzero constants, find the value of ` k .** (**5 marks**)

(b) Evaluate $\int_1^2 (16-7x)^3 dx$ **(5marks)**

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3 (a) Prove that $\frac{\text{Cot}\ \theta}{\text{Csc}\ \theta+1} = \frac{\text{Csc}\ \theta-1}{\text{Cot}\ \theta}$ **(5 marks)**

(b) Solve the equation $cos^2x - \sin x \cos x = 0$ for $x \leq 0 \leq \pi$ **(5 marks)**

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4.(a) The point P (2,10) lies on the curve $y = 3x^2 + 5x - 12$. Find the equation of the normal at the point P. **(6marks)**

(b) The cost of framing a picture is K20 per cm^2 of glass and K50 per cm^2 of wooden frame. A rectangular picture is such that its length is 4 cm greater than its width. Given that the width of the frame is x cm and that a maximum of K10 000 is available for framing, calculate the range of the possible lengths of the width of the frame. **(6 marks)**

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5. The area of a circle increases at the rate of 0.32 cm^2/s , find the rate of increase of radius when the area is 16 cm^2, leaving your answer in terms of π **(6marks)**

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6. Simplify $\frac{1}{\cos x\sqrt{1+\cot^2 x}}$, giving the answer as tangent of an angle. **(6marks)**

7. Show that if $\left|\frac{x+1}{x+2}\right| < 2$, then $(3x + 5)(x + 3) > 0$ **(7marks)**

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Section B (40 marks)

Answer any two questions from this section

8 (a) Given that $f(n) = \frac{2n}{1+n}$ and $f\big(g(n)\big) = 3n - 2$, find $g(1.5)$ **(10marks)**

(b) (i) Using a scale of 2cm to represent 1 unit on the x-axis and 2cm to represent 5units on the y-axis draw, ***on the graph paper provided,*** the graph of $y = 2^{x+1}$ for $-3 \leq x \leq 4$ **(5marks)**.

(ii) Use your graph to find the value of y when $x = 2^{3.8}$ **(5marks).**

9(a) Given that the nth term of the expansion $\left(x - \frac{1}{x} \right)^{17}$ is $\frac{24130}{x}$, find the value of n.

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b. The normal whose equation is $5y + x = 4$ meets the curve $y = x^2 - 3x - 4 \ at \ point \ P$.
Find the equation of the tangent drawn through P. **(10 marks)**

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10 (a) **Figure 1** below shows an area bound by the curve y = f(x) and the x - axis.

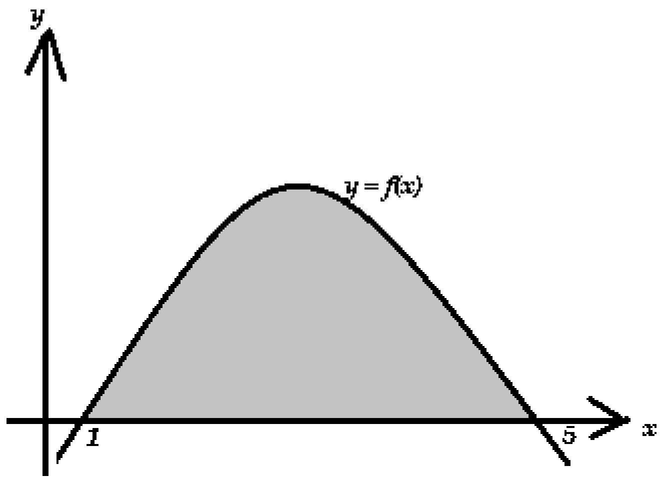

Figure 1

Given that the curve crosses the x – axis at x = 1 and x = 5, find the area of the shaded part.

(10marks)

(b) Using a scale of 2 cm to represent $\frac{\pi}{4}$ radians on the horizontal axis and 5 cm to represent 1 unit on the vertical axis draw, ***on the graph paper provided,***

i. the graph of $y = 1 + 2\cos x$ and $y = \sin x$ for $0 \leq x \leq \pi$.

(8 marks)

ii. Use the graph to solve the equation $1 + 2\cos x = \sin x$

(2 marks)

END OF QUESTION PAPER

NB: This paper contains 12 printed pages.